

MISSOURI S&T
Department of Mathematics and Statistics

Stat 3113 - Section 101
Exam 1

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Student Name: Key

Good luck and happy reading work!

Instructions:

1. Be sure to print your first and last name in the spaces provided at the top of this page.
 2. Work all problems. **Show all work.** Full credit will be given only if work is shown which *fully justified* your answer.
 3. There should be sufficient space under each problem in which to show your work. **Do not use the back of the page where the problem is on.**
 4. There are **7 problems**. Point values are indicated (in parentheses) for each problem. The total number of points is 100.
 5. Turn off your cell phone.
 6. Calculator is allowed in this Exam.
 7. There are 12 pages in this Exam 1 set (including this page). **Please check to make sure you have 12 pages with 7 problems.**
 8. Please ask only your proctor if you have any questions during the exam.
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Distribution of Marks

Question	Points	Score
1	6	
2	12	
3	12	
4	10	
5	20	
6	20	
7	20	
Total:	100	

1. (6 points) **MCQs:** For the multiple choice questions, shade the correct answer.

(a) (2 points) Which of the following terms best describes “when successive observations of a system or phenomenon do not produce exactly the same result.”?

- ☐ Observational Study
☒ Variability

- ☐ Random Experiment
☐ Enumerate Study

(b) (2 points) If $P(A|B) = 0.25$, $P(A \cap B) = 0.32$ and $P(A) = 0.25$, then events A and B are:

- ☐ equally likely events.
☐ independent events.

- ☐ mutually exclusive events.
☒ none of the above.

$$P(B) = \frac{P(A \cap B)}{P(A|B)} = \frac{0.32}{0.25} = 1.28 \text{ (Impossible)}$$

(c) (2 points) A **Poisson distribution** is a reasonable model for the random variables in each of the following **except**

- ☐ number of website visitors per hour.
☐ number of network failures per week.
☒ number of donors who will be tested until a match is found.
☐ number of calls per hour at the admissions office.

2. (12 points) **True/False:** Indicate whether the following statements are true or false.

- (a) ☐ True ☒ False When the sample is used to make conclusion about the population from which the sample was drawn, it's referred to as an Analytic study.
- (b) ☐ True ☒ False The overall median (Q_2) is included in the observations considered to be the lower half.
- (c) ☒ True ☐ False In general, complementary events are mutually exclusive, but the converse is not true.
- (d) ☒ True ☐ False A sample from the population does not necessarily have to share the same characteristics as the population.
- (e) ☐ True ☒ False The range of a discrete random variable can not be countably infinite.
- (f) ☐ True ☒ False The number of trials until the r th success in a Negative Binomial Experiment can be anywhere from 0 to ∞ .

3. (12 points) **IVPPSS:** I have a friend who wanted to start a (fishing) bait store on the West end of Ashland in 2003. He wanted to determine what proportion of Ashland residents that currently used the East end bait store would use a store in the West end if one existed. He sent out 5000 questionnaires and received 2378 back from patrons of the East end store. Use this information to perform an IVPPSS.

Identify the **Individual, Variable, Population, Parameter, Sample and Statistic** in this scenario.

Label	Description
<i>Parameter</i>	Proportion of ALL patrons of the East end store in 2003 that would go to the West end store.
<i>Sample</i>	The 2378 patrons of the East end store in 2003 that returned the questionnaire.
<i>Individual</i>	An Ashland resident that patronized the East end store in 2003.
<i>Statistic</i>	Proportion of the 2378 patrons of the East End store that would go to West end store.
<i>Variable</i>	Whether they would go to the West end store or not.
<i>Population</i>	All Ashland residents that patronized the East end store in 2003.

4. (10 points) **MATCHING:** For problems below, write the letter from the list below that best matches the description. Note: Each answer may be used once, more than once, or not at all.

- | | | |
|-----------------------|------------------------|------------------------|
| A Experiment | E Relative frequency | I Factor |
| B Response | F Geometric experiment | J Retrospective Study |
| C Factorial | G Replication | K Bernoulli experiment |
| D Observational study | H Random variable | L Controller |

(a) *A* The process of obtaining an observed result of some phenomenon.

(b) In a designed experiment, the engineer makes deliberate or purposeful changes in controllable variables called *I* .

(c) A week ago, the home page of an e-commerce site was updated. Purchasing rates are compared between users who saw the old and new home page. This is an example of *D* .

- (d) A random experiment with exactly two possible outcomes, "success" and "failure", in which the probability of success is the same every time the experiment is conducted is called: B
- (e) A H / B is a numerical variable whose measured value can change from one replicate/trial to another.

5. (20 points) **Estimating the Most Likely Location of Oxygen-Cathode Failure.**

Plant operators are attempting to eliminate the most likely cause of failures in plants and also, draw cautionary measures that would be advisable in the design of future plants. As a result, they consider the possible failure of three identical oxygen-cathodes during a particular month of operation.

The failures are stipulated to have occurred in three independently operating electrochemical plants. Out of the 150 combined failures, 39 are from plant 1, 51 are from plant 2 and the reminder are from plant 3. Of the failures recorded from plant 1, 13 of them are due to human error whereas 17 and 24 are number of failures due to human error for plants 2 and 3 respectively.

Use the following events to answer the questions below.

$A_1 = \{\text{Plant 1}\}; \quad A_2 = \{\text{Plant 2}\}; \quad A_3 = \{\text{Plant 3}\};$
 $B = \{\text{Cathode failure due to human error}\}$

Note: Among the many possible causes/sources of cathode failures, only two are considered here: "Human Error" or "Binder Content".

- (a) (3 points) **Compute the probability of failure arising from plant 2 and due to human error.**

$$\begin{array}{l|l} \cdot IP(A_1) = \frac{39}{150} = 0.26 & \cdot IP(B/A_1) = \frac{13}{39} = 0.33 \\ \cdot IP(A_2) = \frac{51}{150} = 0.34 & \cdot IP(B/A_2) = 0.33 \\ \cdot IP(A_3) = \frac{60}{150} = 0.4 & \cdot IP(B/A_3) = 0.4 \end{array}$$

Here, we want

$$\begin{aligned} IP(A_2 \cap B) &= IP(B/A_2) \cdot IP(A_2) \\ &= 0.33 \times 0.34 \end{aligned}$$

Answer =

0.11

- (b) (4 points) **Determine the probability of failure arising from human error. What probability rule is used here?**

By the law of total probability, we have:

$$\begin{aligned} IP(B) &= IP(B/A_1) \cdot IP(A_1) + IP(B/A_2) \cdot IP(A_2) + IP(B/A_3) \cdot IP(A_3) \\ &= (0.33 \times 0.26) + (0.33 \times 0.34) + (0.40 \times 0.40) \end{aligned}$$

Answer =

0.36

- (c) (3 points) Determine the probability of failure arising from binder content.

$$\begin{aligned} IP(B') &= 1 - IP(B) \\ &= 1 - 0.36 \end{aligned}$$

Answer =

0.64

- (d) (7 points) Given that the binder content is the cause of failure, which plant is most likely to fail.

Hint: Use the Bayes's rule to find the three probabilities and compare them.

$$\begin{aligned} IP(A_i | B') &= \frac{IP(A_i \cap B')}{IP(B')} \\ &= \frac{IP(A_i) - IP(A_i \cap B)}{IP(B')} \\ &= \frac{IP(A_i) [1 - IP(B | A_i)]}{IP(B')} \quad , \quad i = 1, 2, 3 \end{aligned}$$

$$\bullet IP(A_1 | B') = \frac{0.26 [1 - 0.33]}{0.64} = 0.27$$

$$\bullet IP(A_2 | B') = \frac{0.34 [1 - 0.33]}{0.64} = 0.35$$

$$\bullet IP(A_3 | B') = \frac{0.40 [1 - 0.40]}{0.64} = \underline{\underline{0.38}} \text{ (highest)}$$

Answer =

Plant 3.

- (e) (3 points) Find the probability that plant 1 does not fail given that binder content is considered to be the cause of failure.

$$\begin{aligned} P(A'|B') &= 1 - P(A,|B') \\ &= 1 - 0.27 \end{aligned}$$

Answer = 0.73

6. (20 points) Truck tires are tested over rough terrain. 4 trucks go through the test. Assume that the blowouts are random and independent. The table below gives the distribution of the number of trucks that fail to complete the test run due to blowout.

Number of trucks that have blowouts (X)	0	1	2	3	4
$f(x) = P(X = x)$	x	0.12	0.11	0.33	y

- (a) (3 points) Given that on average, we expect 2 trucks to have blowouts, that's $E(X) = 2$, find the value of y .

$$E[X] = \sum_{\text{all } x} x f(x)$$

$$(0)(x) + (1)(0.12) + (2)(0.11) + (3)(0.33) + (4)(y) = 2$$

$$\Rightarrow 1.33 + 4y = 2 \Rightarrow 4y = 0.67$$

Answer =

$$y = 0.1675$$

- (b) (3 points) Now, find the value of x that makes the table a valid probability mass function (pmf).

$$\sum_{\text{all } x} f(x) = 1 \Rightarrow x + 0.12 + 0.11 + 0.33 + 0.1675 = 1$$

$$\Rightarrow x + 0.7275 = 1$$

Answer =

$$x = 0.2725$$

- (c) (4 points) Compute the probability that fewer than three trucks have blowouts.

$$\begin{aligned} P(X < 3) &= 1 - P(X \geq 3) \\ &= 1 - [f(3) + f(4)] \\ &= 1 - [0.33 + 0.1675] \end{aligned}$$

Answer =

$$0.5025$$

- (d) (4 points) Suppose that it costs $\$(100 - X)^2$ to replace a tire. Find the expected cost needed to be spent.

$$E[(100 - X)^2] = \sum_{\text{all } x} (100 - x)^2 f(x)$$

$$\begin{aligned} &= (100 - 0)^2 \times 0.2725 + (100 - 1)^2 \times 0.12 + (100 - 2)^2 \times 0.11 \\ &\quad + (100 - 3)^2 \times 0.33 + (100 - 4)^2 \times 0.1675 \end{aligned}$$

Answer =

$$\$9606.21$$

7. (20 points) In manufacturing integrated circuits, the yield of the manufacturing process is the percentage of good chips produced by the process. The probability that an integrated circuit manufactured by Ace Electronics Company will be defective is $p = 0.05$. A random sample of 15 circuits is selected for the testing.

Clearly show work for parts b,c and d.

- (a) (4 points) Let the random variable X be the number of defective integrated circuits in the sample. Determine its distribution.

Hint: The pmf and its domain.

$$X \sim \text{Bin}(15, 0.05), \quad X = 0, 1, \dots, 15.$$

$$f(x) = \binom{15}{x} (0.05)^x (1 - 0.05)^{15-x}$$

Answer =

- (b) (3 points) Compute the probability that no more than two of the integrated circuits will be defective in the sample. 4 decimal places.

$$\begin{aligned} P(X \leq 2) &= f(0) + f(1) + f(2) \\ &= \binom{15}{0} (0.05)^0 (1 - 0.05)^{15} + \binom{15}{1} (0.05)^1 (1 - 0.05)^{14} \\ &\quad + \binom{15}{2} (0.05)^2 (1 - 0.05)^{13} \end{aligned}$$

Answer =

0.9638

- (c) (3 points) Find the probability that there will be at least one of defective integrated circuit within the sample.

$$\begin{aligned} P(X \geq 1) &= 1 - P(X < 1) \\ &= 1 - f(0) \\ &= 1 - \binom{15}{0} (0.05)^0 (1 - 0.05)^{15} \end{aligned}$$

Answer =

0.5367

- (d) (10 points) Define the random variable Y as **the number of integrated circuits selected until 5 defective circuits are found during testing**. Use the same value of p to answer the questions below.

(i) (3 points) **What distribution does Y follow?**

$$Y \sim NB(r=5, p=0.05)$$

$$f(x) = \binom{x-1}{4} (1-0.05)^{x-5} (0.05)^5, \quad x = 5, 6, \dots$$

Answer =

- (ii) (3 points) **How far, roughly on average, will the number of integrated circuits selected until 5 defective circuits are found during testing, deviate from the mean?**

$$V(Y) = \frac{r(1-p)}{p^2} = \frac{5(1-0.05)}{(0.05)^2} = 1900.$$

$$\Rightarrow \sigma = \sqrt{\text{Var}(Y)} = \sqrt{1900}$$

Answer =

43.59

- (iii) (4 points) **How many integrated circuits do you expect to select during testing until ONE defective circuit is found?**

$$X \sim \text{GEO}(p=0.05)$$

$$E[X] = \frac{1}{p} = \frac{1}{0.05}$$

Answer =

20

Formula Sheet

$$1. Q_1 - (1.5 \cdot IQR) \qquad Q_3 + (1.5 \cdot IQR)$$

$$2. (a + b)^2 = a^2 + 2ab + b^2$$

$$3. P(A') = 1 - P(A)$$

$$4. P(A|B) = \frac{P(A \cap B)}{P(B)}, \text{ provided } P(B) > 0$$

$$5. P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$6. P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

$$7. P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)$$

8. If A_1, A_2, A_3 are 3 mutually exclusive and exhaustive events and B is any event,

$$P(A_1|B) = \frac{P(B|A_1)P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)}$$

Note: This generalizes to k mutually exclusive and exhaustive events

9. $P(A \cap B) = P(A) \cdot P(B)$ if A, B are independent.

10.

$$\mu = E(X) = \sum x f(x)$$

11.

$$\text{Var}(X) = E(X^2) - \mu^2$$

12.

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

$$E(X) = np \quad \text{and} \quad \text{Var}(X) = np(1-p)$$

13.

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

$$E(X) = \lambda \quad \text{and} \quad \text{Var}(X) = \lambda$$

14.

$$f(x) = (1-p)^{x-1} p, \quad x = 1, 2, \dots$$

$$E(X) = \frac{1}{p} \quad \text{and} \quad \text{Var}(X) = \frac{(1-p)}{p^2}$$

15.

$$f(x) = \binom{x-1}{r-1} (1-p)^{x-r} p^r, \quad x = r, r+1, r+2, \dots$$

$$E(X) = \frac{r}{p} \quad \text{and} \quad \text{Var}(X) = \frac{r(1-p)}{p^2}$$